

M Charity (they/them)

Assistant Professor (University of Richmond) - Game Design and AI Researcher
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Education

New York University - Tandon School of Engineering (Class of 2024)

Brooklyn, NY

PhD, Game Design and Artificial Intelligence, GPA 3.809/4.00

Aug 2019 - Apr 2024

- 2022-2024 GAANN Scholar Recipient
- 2021 NSF GRFP Runner-up Recipient
- Thesis: Online Creative Collaborative Content Generation

Virginia Commonwealth University - College of Engineering (Class of 2019)

Richmond, VA

BS, Computer Science with minor in Mathematics, GPA 3.81/4.00

Aug 2016 - May 2019

- 2019 College of Engineering Outstanding Student Award
- 2018 Wright Access Scholarship Recipient
- 2017 and 2018 EDU Tech Scholarship Recipient
- 2016 Provost Scholarship Recipient

Deep Run High School - Center for Information Technology (Class of 2016)

Glen Allen, VA

High School Diploma, GPA 4.94/4.00 (weighted)

Sept 2012 - Jun 2016

- 2015 NCWIT National Aspiring Award Winner
- 2014 NCWIT Regional Aspiring Award Winner

Research Experience

University of Richmond

Richmond, VA

Assistant Professor and Researcher

Jun 2024 - *present*

- Studying game design and artificial intelligence and collaborative tool design
- Principal investigator for multiple research projects and collaborations internally and externally
- Recent Topics: game search engines, artificial agent design, MMO game design, simulation design

New York University - Tandon School of Engineering

Brooklyn, NY

PhD Researcher

Sep 2019 - Jun 2024

- Thesis: Online Creative Collaborative Content Generation [[video link](#) | [doc link](#)]
- Studying game design and artificial intelligence under Dr. Julian Togelius
- Designed, developed, and published a range of online creative content generation tools featuring technologies such as general game-playing agents, quality diversity and evolutionary algorithms, large-language models, open-ended frameworks, and abstract simulation environments
- Led multiple teams of undergraduate and masters students in game AI research projects

Virginia Commonwealth University

Richmond, VA

Unity Programmer MCV moVR - Virtual Reality Physical Therapy Lab

Jan 2019 - Aug 2019

- Built 5-game virtual reality game suite for fibromyalgia patients (Build-A-Bot)
- Prototyped HTC Vive full-body motion capture system for animation
- Built pain rating system for viewing work-effort of physical activities
- Built card matching VR game for fibromyalgia patients
- Prototyped pain categorization and selection VR system

Natural Language Processing Undergraduate Researcher

Jun 2016 - Jan 2019

- Developed named-entity recognition system for drug occurrences
- Integrated interface for SemMed Database
- Helped develop literature based discovery neural network
- Developed latent semantic analysis system for doctor-patient interaction

University of Minnesota - Twin Cities

Minneapolis, MN

NSF Big Data REU Researcher

Jun 2018 - Aug 2018

- Worked in Dr. Victoria Interrante's virtual reality lab
- Designed and implemented a novel locomotion VR system
- Ran a user-based experiment with 10 participants for the performance of the system
- Work was featured in VCU Engineering news: <https://egr.vcu.edu/news-events/news/archive/vr-skateboard.html>

Teaching Experience

University of Richmond

Richmond, VA

Assistant Professor

Aug 2025 - *present*

- CMSC 150 - Introduction to Computing w/ Lab (Fall 2025/Spring 2026; 2 sections)
- CMSC 240 - Software Systems and Development w/ Lab (Fall 2025/Spring 2026; 2 sections)
- Mentored 4 undergraduate students during fall and spring semesters

Visiting Assistant Professor

Aug 2024 - Aug 2025

- CMSC 150 - Introduction to Computing w/ Lab (Fall 2024/Spring 2025; 4 sections)
- CMSC 395 - Special Topics - Game Design (Spring 2025; 1 section)
- Mentored 2 undergraduate students during Summer 2025 on research projects

New York University

Brooklyn, NY

Adjunct Professor

Jan - May 2023, 2024

- Course: CS-GY 6553 / CS-UY 4553 (Undergraduate/Graduate Game design)
- Created and taught lectures and ran administrative tasks such as entering grades
- Answered student questions and aided with game engine technical difficulties
- Critiqued and reviewed prototype games during class presentations
- Taught Spring semester 2023 and 2024

Teaching Assistant

Brooklyn, NY

Aug 2020 - Dec 2023

- Teaching assistant for multiple CS graduate and undergraduate classes
- Artificial Intelligence (CS-GY 6613) - Fall semester 2020 and 2021
- AI for Games (CS-GY 6943) - Spring semester 2021
- Game Design (CS-UY 4553, CS-GY - 6553) - Spring semester 2022 and Fall semester 2023
- Graded projects, wrote homework assignments, and tutored during office hours

Virginia Commonwealth University

Richmond, VA

Teaching Assistant - CMSC 257

Aug 2018 - May 2019

- Teaching assistant grading projects, aiding with lab assignments, and tutoring during office hours

CodeVA

Richmond, VA

Teaching Assistant - Game Design Summer Camp

Aug 2015 - Sep 2015

- Assisted with teaching Scratch and game development to elementary school children

Technical Work Experience

modl.ai

Research Consultant Contractor

Copenhagen, Denmark (Remote)

- Researched and developed the flexible replay prototype system for the startup modl.ai Jul 2023 - Sep 2023
- Used the Unity game engine to develop a regression testing game environment
- Designed and developed a local search-based agent to replay and adapt to user playtraces of levels

LAIKA Team

Research Consultant Contractor

Copenhagen, Denmark (Remote)

- Researched and developed the Character Creator tool for the startup company LAIKA May 2022 - Aug 2022
- Researched up-and-coming NLP technologies to implement in the LAIKA product

University of Richmond

Web Services Intern

Richmond, VA

Jun 2015 - Aug 2015

- Intern to the back-end web services team monitoring statistics for web traffic
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Service

IEEE Conference on Games

Co-track chair for Procedural Content Generation track 2026

Madrid, Spain

Mar 2026 - *present*

- Make recommendations of acceptance

AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment

Co-track chair for Demo track 2025

Alberta, Canada

Mar 2025 - Nov 2025

- Co-wrote CfP, reviewed submissions, and managed author communications
- Co-chaired the workshop on the day of the event

Co-organizer for EXAG 2021 Workshop

Virtual

Mar 2021 - Oct 2021

- Helped organize and host the 2021 Experimental AI in Games Workshop at AIIDE 2021
- Maintained and updated the EXAG.org website
- Put together the Call for Papers, arranged presentation schedule, and recruited reviewers

Foundations of Digital Games

Co-organizer for PCG Workshop 2024

Worcester, MA

Sept 2023 - May 2024

- Helped organize the 2024 Procedural Content Generation Workshop at FDG 2024
- Co-wrote and submitted the workshop proposal document
- Co-chaired the workshop on the day of the event

FDGTV Producer / Presenter

Lisbon, Portugal

Apr 2023

- Interviewed multiple academic authors and presenters at the FDG 2023 conference
- Interviews were posted on the FDGTV YouTube channel

Dagstuhl Seminar 25292

Co-organizer for Dagstuhl Seminar 25292

Wadern, Germany

Mar 2024 - present

- Seminar Topic: New Frontiers in AI for Game Design
- Co-wrote proposal and description for the seminar
- Helped with arranging invite list for attendants
- Further administrative tasks to be done closer to the event

Fostering Futures Forum - Technology Leadership Circle

Community Gaming Working Group Technology Specialist/Faculty

Virtual + Boston, MA

Dec 2024 - Nov 2025

- Make recommendations of acceptance
- Met monthly virtually for 2 hours to design a prototype gaming product for foster youth
- Presented design plans at Fostering Futures Forum hosted by MIT

Deep Run High School

CIT Advisory Council member

Glen Allen, VA

Aug 2025 - *present*

- Volunteer as a member of the CIT program advisory council for DRHS
- Meet every month to discuss CIT events with students and provide council from academic perspective

CIT Projects Client

Jan 2026 - *present*

- Acting client for sophomore-junior level group project for 6-9 CIT students
- Project #1: Meme Search Engine
- Project #2: AI Generated Slideshow Karaoke Generator

CIT Capstone Mentor

Aug 2024 - *present*

- Advising senior CIT students for the development of a capstone project
- 2024-25: Kevin Matthew - Developing a chatbot for the Richmond Flying Squirrels
- 2025-26: Emi Brenigar - Developing a 2048-style puzzle game

University of Richmond

Computer Science Department Tenure Track Hiring Committee

Richmond, VA

Sep 2025 - present

- Reviewing and evaluating faculty applications
- Conducting virtual and on-campus interviews with candidates

Virginia Commonwealth University

Computer Science Student Advisory Board member

Richmond, VA

May 2018 - May 2019

- Communicated with both students and professors about state of the department
- Volunteer as ambassador at College of Engineering open houses
- Promote student-led social clubs and extracurricular activities

Invited Talks, Conferences, and Workshops

Virginia Commonwealth University 2026

Richmond, VA

Jan 2026

- Invited to give a seminar talk for the VCU Seminar Series
- Talk Title: Made by Me and ChatGPT: Using AI as a Collaborative Partner for Creative Tasks

Virginia Commonwealth University 2025

Richmond, VA

Feb 2025

- Invited to give a seminar talk for the VCU Seminar Series
- Talk Title: Friend vs. Foe: Evaluating AI Assisted Content Generation for Game Design

Midlothian High School 2025

Midlothian, VA

Feb 2025, Apr 2025

- Invited to give a seminar talk to the AI club
- Talk Title (Feb): HCI in Game AI Tools
- Talk Title (Apr): Procedural Content Generation in Games

Deep Run High School Computer Science Honor Society 2024

- Invited as a guest speaker for the DRHS Computer Science Honor Society induction ceremony
- Talk Title: My Journey as a Computer Scientist thus Far

Glen Allen, VA
Sep 2024

Schloss Dagstuhl Seminar 2024

- Invited to attend Schloss Dagstuhl Seminar 2024
- Seminar Topic: Computational Creativity for Game Development

Wadern, Germany
Jun 2024

King's College 2024

- Invited to give a seminar talk to King's College by Dr. Michael Cook
- Seminar Topic: AI, Video Games, and the Web:

London, GB, United Kingdom
Jun 2024

MIT / Foster America Youth Tech Workshop 2023

- Invited panelist for the first MIT / Foster America Youth Tech Workshop
- Talked about research work and life story to foster youth ages 16-25
- Discuss generative AI and brainstormed app product design in break-out groups

New York, NY
Dec 2023

ALOE Workshop 2023

- Invited speaker for the Agent Learning in Open-Ended Environments Workshop
- Workshop during the NeurIPS 2023 conference in New Orleans, Louisiana
- Talk Title: Amorphous Fortress: Exploring Emergent Behavior in Open-Ended Simulations

New Orleans, LO
Dec 2023

Games for Change 2023

- Invited speaker for the Games for Change conference in New York City, New York
- Talk Title: Using AI in Game Design

New York City, NY
Jul 2023

PWLConf 2022

- Invited speaker for the Papers We Love conference in St Louis, Missouri
- Talk Title: 3CG: Collaborative and Creative Content Generation in Game Design

St. Louis, MO
Sep 2022

Game Devs of Color 2022

- Invited speaker for the Game Devs of Color conference (virtual)
- Talk Title: Developing Cooler PCG Systems based on Real-Time Player Input

Virtual
Sep 2022

Leadership/Teamwork Experience

FRC Team 1086 - Blue Cheese

- Lead Programmer
- FTC Captain Team 93

Glen Allen, VA
May 2013 - Jul 2015
Jul 2015 - Feb 2016

Deep Run High School

- Founder / Leader of Computer Club Game Design Division
- CyberPatriot (JV Captain)

Glen Allen, VA
Oct 2012 - May 2015
Oct 2012 - May 2015

Technical Skills

Programming Languages

- Javascript, Python, PHP, MySQL, C#, C++, C, Java, Perl

Software, game engines, and operating systems

- HTML5, AWS, Unity, Godot, PICO-8, Bitsy, NodeJS, Google Sheets, git, jupyter-notebook, unix, MacOS, Windows
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Publications

Journal Publications

- **Charity, M.**, Dave, I., Khalifa, A., & Togelius, J. (2022, November). Baba is Y'all 2.0: Design and Investigation of a Collaborative Mixed-Initiative System. *IEEE Transactions on Games*.

Conference Publications

- Szajda, B., & **Charity, M.** (2025, November). GameGoogle: A Search Engine for Mechanics in Video Games. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment* (Vol. 21, No. 1, pp. 406-410).
- **Charity, M.**, Earle, S., Rajesh, D., Wilson, M., & Togelius, J. (2024, July) Amorphous Fortress: Observing Emergent Behavior in Multi-Agent FSMs. In the *Proceedings of the World Congress on Computational Intelligence*, 2024.
- Anjum, A., Li, Y., Law, N., **Charity, M.**, & Togelius, J. (2024, May) The Ink Splotch Effect: A Case Study on ChatGPT as a Co-Creative Game Designer. In the *Proceedings of the 19th International Conference on the Foundations of Digital Games*.
- Merino, T., Negri, R., Rajesh, D., **Charity, M.**, & Togelius, J. (2023, October). The Five-Dollar Model: Generating Game Maps and Sprites from Sentence Embeddings. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment* (Vol. 19, No. 1, pp. 107-115).
- Green, M. C., Khalifa, A., **Charity, M.**, & Togelius, J. (2022, September). Persona-driven Dominant/Submissive Map (PDSM) Generation for Tutorials. In *Proceedings of the 17th International Conference on the Foundations of Digital Games* (pp. 1-10).
- **Charity, M.**, & Togelius, J. (2022, October). Aesthetic Bot: Interactively Evolving Game Maps on Twitter. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment* (Vol. 18, No. 1, pp. 18-25).
- **Charity, M.**, Memon, N., Jiang, Z., Sen, A., & Togelius, J. (2022, September). Diversity and Novelty MasterPrints: Generating Multiple DeepMasterPrints for Increased User Coverage. In *2022 International Conference of the Biometrics Special Interest Group (BIOSIG)* (pp. 1-4). IEEE.
- Green, M. C., Khalifa, A., **Charity, M.**, Bhaumik, D., & Togelius, J. (2022, July). Predicting Personas Using Mechanic Frequencies and Game State Traces. In *2022 IEEE Congress on Evolutionary Computation (CEC)* (pp. 1-8). IEEE.
- Jaramillo, C., **M. Charity**, R. Canaan, and J. Togelius. (2020, October) Word Autobots: Using Transformers for Word Association in the Game Codenames. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 16, no. 1, Oct. 2020, pp. 231-7
- **Charity, M.**, D. Rajesh, R. Ombok, and L. B. Soros. (2020, October) Say 'Sul Sull' to SimSim, A Sims-Inspired Platform for Sandbox Game AI. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 16, no. 1, Oct. 2020, pp. 182-8
- **Charity, M.**, Khalifa, A., & Togelius, J. (2020, July). Baba is Y'all: Collaborative Mixed-Initiative Level Design. *2020 IEEE Conference on Games (CoG)*, 542-549.
- **Charity, M.**, Green, M. C., Khalifa, A., & Togelius, J. (2020, September). Mech-elites: Illuminating the mechanic space of gvg-ai. In *International Conference on the Foundations of Digital Games* (pp. 1-10).

Workshop Publications

- Rajesh, D., Togelius, J., & **Charity, M.** (2025, November). Amorphous Interpretations: Diverse Narrative Generation for Open-Ended Simulation Environments. In *Proceedings of the Interactive Narratives Workshop*
- Earle, S., **Charity, M.**, Rajesh, D., Wilson, M., & Togelius, J. (2023, December). Quality Diversity in the Amorphous Fortress : Evolving for Complexity 0-Player Games. Accepted to ALOE 2023.
- **Charity, M.**, Bhartiya, Y., Zhang, D., Khalifa, A., & Togelius, J. (2023, August). A Preliminary Study on a Conceptual Game Feature Generation and Recommendation System. In the *Proceedings of Conference on Games 2023*.
- **Charity, M.**, Rajesh, D., Earle, S., & Togelius, J. (2023, July). Amorphous Fortress: Observing Emergent Behavior in Multi-Agent FSMs. *arXiv preprint arXiv:2306.13169*. Artificial Life Conference - ALIFE for and from Games Workshop, 2023.

- Merino, T., **Charity, M.**, & Togelius, J (2023, April). Interactive Latent Variable Evolution for the Generation of Minecraft Structures. International Conference on the Foundations of Digital Games Workshop on Procedural Content Generation.
- **Charity, M.**, & Togelius, J. (2022, August). Keke AI Competition: Solving puzzle levels in a dynamically changing mechanic space.